ImmersionTrack

Grade 12 Work Immersion Management System

1. System Overview

ImmersionTrack is a Django-based web application designed to manage Grade 12 work immersion programs. It provides a centralized platform for students, company supervisors, and the school dean to coordinate placements, attendance, task submissions, daily journals, evaluations, announcements, and reports.

The system supports:

- Role-based access for students, supervisors, and the dean
- Student onboarding, approval workflows, and company placement
- Task management, submissions, grading, and feedback
- Attendance tracking with multi-phase QR code support
- Daily journal/logbook entry review and approval
- Supervisor evaluations with rubric-based scoring
- Announcements and in-app notifications
- Audit logs and administrative monitoring

1.1 What's New in the System

- New multi-phase attendance QR workflow for time-in, break-out, break-in, and time-out tracking
- Student application approval and automated group assignment tracking
- Supervisor task creation and submission review with late/returned statuses
- Daily journal submission review by supervisors with approval/rejection workflow
- Rubric-based evaluation with final score computation and grade generation
- Company join codes for secure student enrollment into immersion groups
- Attendance backups and downloadable records for compliance and audit
- Custom dashboard summaries for dean, supervisor, and student roles

2. Core Objectives

- Simplify work immersion administration for Grade 12 programs
- Replace manual spreadsheet tracking with digital workflows
- Improve communication between students, supervisors, and school administrators
- Generate accurate performance, attendance, and completion reports
- Maintain accountability through audit trails and approval workflows

3. Technology Stack

- Django 4.2+ web framework
- SQLite by default for development (PostgreSQL recommended for production)
- Bootstrap 5 for responsive interface components
- Django Jazzmin for a polished administration UI
- ReportLab for PDF reports
- OpenPyXL for Excel exports
- QR code generation for attendance tracking
- OpenCV / Pyzbar support for scanning codes and verifying attendance data

4. User Roles & Access

Student

- Register for immersion and submit profile details
- Await approval from the dean
- Join a company or immersion group using a join code
- Submit task files and receive supervisor feedback
- Log daily journals and reflections
- View attendance records, task grades, and evaluations
- Submit leave or excuse requests with supporting documents

Supervisor

- Account created by the dean; cannot self-register
- Manage assigned immersion group(s) and company students
- Publish tasks and set deadlines
- Review and grade student task submissions
- Verify attendance through time-in, break, and time-out records
- Review daily journals and approve or reject entries
- Complete rubric-based evaluations for assigned students
- Post announcements to group or school-wide audiences

Dean / Administrator

- Full administrative oversight of the immersion system
- Approve or reject student registration and placement requests
- Create companies, immersion groups, and join codes
- Add and manage supervisors

- Monitor attendance, journal entries, reports, and student progress
- View audit logs for system operations
- Manage announcements and notifications
- Download attendance backups and administrative reports

5. Key Modules

Accounts

- Custom `User` model with role selection (`student`, `supervisor`, `dean`)
- Student registration with profile creation and approval workflow
- Profile pictures, account activation, and role-specific permissions

Students

- `StudentProfile` extends user data with student ID, strand, section, school year, contact details, and application status
- Leave requests and requirement uploads are tracked in this module
- Students can only access their own records via role-based decorators

Companies

- `Company` record includes industry type, contact details, required hours, and join code
- `ImmersionGroup` defines the work immersion group, company, supervisor, and date range
- `StudentCompanyAssignment` joins students to groups and tracks assignment status
- Join codes allow students to enroll in groups securely

Tasks

- Supervisors create `Task` records with title, description, instructions, due date, score, and attachments
- Students submit `TaskSubmission` files, comments, and later receive scores and feedback
- Submission statuses include submitted, late, reviewed, graded, and returned for revision

Attendance

- Multi-phase attendance tracking with `AttendanceRecord` fields for time-in, break-out, break-in, and time-out
- QR code support via `AttendanceQR` for secure attendance capture
- Records include hours rendered, attendance status, approval status, verifier, and remarks
- Students submit excuse requests when absent or unable to report
- Attendance backup files are available for download

Journals

- Students submit `DailyJournal` entries describing daily activities, skills learned, challenges, and reflections

- Supervisors review journals and provide remarks
- Entry status is tracked as pending, approved, or rejected

Evaluations

- Supervisors rate students across seven criteria: punctuality, attendance, work quality, professionalism, communication, teamwork, initiative
- `Evaluation` computes a combined final score and can derive a grade letter
- Evaluation comments and recommendations are stored for school review

Announcements & Notifications

- Announcements can target all users, only students, only supervisors, or a specific immersion group
- Notifications support event types like task creation, submission updates, attendance alerts, journal reviews, evaluations, and approvals
- Users receive real-time alerts in the system dashboard

Audit Logs

- All critical actions are recorded in `AuditLog`
- Tracks who performed each action, when, and what object was changed
- Includes action type, model name, object detail, and IP address when available
- Helps administrators maintain accountability and trace changes

6. Core System Workflows

Student Onboarding

1. Student registers and submits profile details
2. Dean reviews and approves application
3. Dean assigns the student to a company/group or student joins via company join code
4. Supervisor begins monitoring student progress and attendance

Task Submission Process

1. Supervisor publishes a task with instructions and deadline
2. Student uploads work and submits it through the portal
3. Supervisor reviews submission, adds feedback, and assigns a score
4. Student can view graded results, comments, and resubmit if returned

Attendance and Leave Management

1. Supervisor generates attendance QR codes for time-in, break, and time-out
2. Student scans or records attendance events

3. Attendance record calculates total hours rendered automatically

4. Supervisor approves or rejects attendance and excuse requests

Evaluation and Reporting

1. Supervisor completes rubric evaluation once immersion activities progress

2. System computes a final score and outputs a grade letter

3. Dean can view summary reports of student performance and attendance

4. Reports can be exported for administrative review and presentations

7. System Architecture & Security

- Built on Django with apps separated by function for maintainability
- Custom role-based access using decorators in `core/decorators.py`
- Professional admin UI powered by Jazzmin for improved usability
- Production recommendation: use PostgreSQL, secure email backend, HTTPS, and change default secrets
- Default setup script creates demo accounts but must be replaced with strong credentials before deployment

8. Setup Summary

- Python 3.10+ required
- Install dependencies from `requirements.txt`
- Run migrations for each app module and `python manage.py migrate`
- Optionally use `setup.sh` to create demo users and collect static files
- Start server with `python manage.py runserver 0.0.0.0:8000`

9. Default Demo Credentials

Role Username Password Intended Use

System Administrator / Dean `admin` / `dean` `admin1234` / `dean1234` Admin panel and dean dashboard

Supervisor `supervisor1` `super1234` Supervisor dashboard workflow

Student `student1` `student1234` Student portal workflow

> Note: Change default passwords immediately for production.

10. Recommended Presentation Outline

1. Project name and purpose

2. Target users and roles

3. System architecture and technology stack
4. Main features by module
5. Typical student/supervisor/dean workflows
6. Security and deployment notes
7. Live demo or screenshot examples
8. Benefits for schools and companies

11. Benefits and Impact

- Promotes consistent documentation of student work immersion
- Reduces manual tracking errors for attendance and evaluations
- Improves supervisor feedback and grade transparency
- Enables dean-level oversight through centralized dashboards
- Provides digital records for compliance, auditing, and reporting

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ImmersionTrack is designed to be a complete digital solution for schools and partner companies managing Grade 12 immersion programs. This report can be used directly for presentation slides, handouts, or executive summaries.